

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

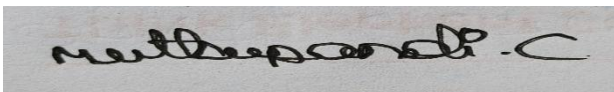
Annexure B
Analytical Problem Solving

<i>Criteria</i>	Problem Understanding (2)	Method Selection (2)	Solution Process (2.5)	Accuracy of Calculation (2)	Interpretation of Results (1.5)	<i>Total(10)</i>
<i>Weightage</i>	20%	20%	25%	20%	15%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I __192565037 / Muthupandi C__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

25/7/20

CGA - 1603

1. Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 36, 40, 45, 46, 52, 70.

- (a) what is the mean of the data? what is the Median?
- (b) what is the Mode of the data? comment on the data's modality (i.e., bimodal, trimodal, etc).
- (c) what is the midrange of the data?
- (d) can you find (roughly) the first quartile (Q_1) & the third quartile (Q_3) of the data?
- (e) Give the five-number summary of the data
- (f) show a boxplot of the data.

(a) Mean & Median

$$\text{Mean} = \frac{\sum x}{n}$$

$$= \frac{809}{27}$$

$$= 29.96 \text{ (approx 30)}$$

Median =

Since $n = 27$ (odd)

$$\text{Median} = \frac{27+1}{2}$$

$$= 14$$

The 14th value = 25

$$\text{Median} = 25$$

(b) Mode & Modality

* 25 appears 4 times

* 35 appears 4 times

$$\text{Mode} = 25 \neq 35$$

Since there are two modes, the data is bimodal.

$$\begin{aligned} \text{(c) Midrange} &= \frac{\text{Min} + \text{Max}}{2} \\ &= \frac{13 + 70}{2} \\ &= \frac{83}{2} \end{aligned}$$

$$\text{Midrange} = 41.5$$

(d) First quartile (Q_1) ~~≠~~ ~~Third~~ quartile (Q_3)
Lower half (Q_1)

13, 15, 16, 16, 19, 20, 20, 21, 22, 22,

25, 25, 25

$$Q_1 = 20$$

$$\begin{aligned} \text{Median of } Q_1 &= \frac{13+19}{2} \\ &= 16 \end{aligned}$$

Upper half (Q_3)

30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46

$$Q_3 = 35$$

$$\text{Median} = \frac{13+1}{2}$$

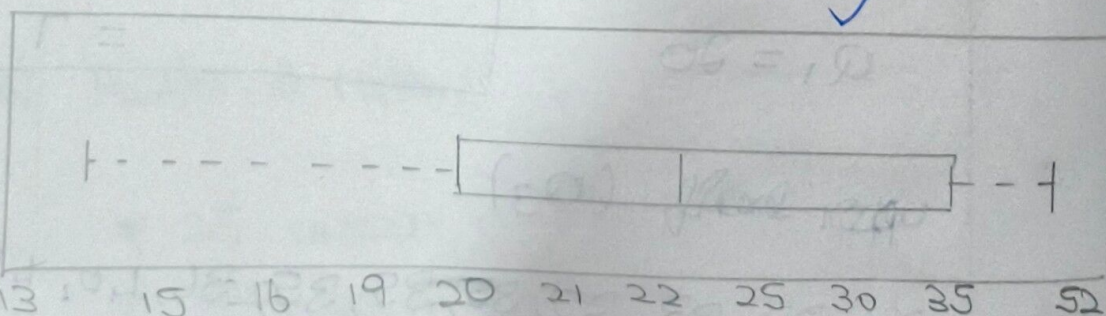
$$= 7$$

(e) Five-Number Summary

- Min = 13
- Q_1 = 20
- Median = 25
- Q_3 = 35
- Max = 70

Statistic	Value
Min	13
Q_1	20
Med	25
Q_3	35
Max	70

(f) Boxplot



2. Given the following measurement for the variable age:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28,

standardize the variable by the following: compute the mean absolute deviation of age.

Mean:

$$\frac{18+22+25+42+28+43+33+35+56+28}{10}$$

$$= \frac{330}{10}$$

$$= 33$$

Absolute deviation Mean;

$$| \text{Age}(x) | - | x - 33 |$$

$$| 18 | - | 15 |$$

$$| 22 | - | 11 |$$

$$| 25 | - | 8 |$$

$$| 42 | - | 9 |$$

128|5|

143|10|

133|0|

135|2|

156|23|

128|5|

absolute deviation;

$$15 + 11 + 8 + 9 + 5 + 10 + 0 + 2 + 23 + 5$$

$$MAD = \frac{88}{10}$$

$$= 8.8$$

3. Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. Use smoothing by bin means to smooth the above data, using

smooth by Bin Means (Bin depth = 3)

There are 27 values.

Since Bin depth = 3, each bin contains 3 values

Bin	values
B ₁	13, 15, 16
B ₂	16, 19, 20
B ₃	20, 21, 22
B ₄	22, 25, 25
B ₅	25, 25, 30
B ₆	33, 33, 35
B ₇	35, 35, 35
B ₈	36, 40, 45
B ₉	46, 52, 70

compute the Mean of each Bin

Bin	values	Mean
B ₁	13, 15, 16	14.67
B ₂	16, 19, 20	18.33
B ₃	20, 21, 22	21
B ₄	22, 25, 25	24
B ₅	25, 25, 30	26.67
B ₆	33, 33, 35	33.67
B ₇	35, 35, 35	35
B ₈	36, 40, 45	40.33
B ₉	46, 52, 70	56

Replace Bin with Bin Mean

Bin	Smooth values
B ₁	14.67, 14.67, 14.67
B ₂	18.33, 18.33, 18.33
B ₃	21, 21, 21
B ₄	24, 24, 24
B ₅	26.67, 26.67, 26.67
B ₆	33.67, 33.67, 33.67
B ₇	35, 35, 35
B ₈	40.33, 40.33, 40.33
B ₉	56, 56, 56

4. use the two method below to normalize the following group of data 200, 300, 400, 600, 1000 (a) Min-Max normalization by setting min = 0 & max = 1 (b) z-score normalization.

200, 300, 400, 600, 1000

(a) min-max normalization (min = 0, max = 1)

$$\text{new value} = \frac{x - \text{min}}{\text{max} - \text{min}}$$

• Min value = 200

• Max value = 1000

$$\text{Range} = 1000 - 200$$

$$= 800$$

original value	calculation	normalized value
200	$(200 - 200) / 800$	0.00
300	$(300 - 200) / 800$	0.125
400	$(400 - 200) / 800$	0.250
600	$(600 - 200) / 800$	0.500
1000	$(1000 - 200) / 800$	1.000

(b) Z-Score

$$\begin{aligned}\text{Mean} &= \frac{200 + 300 + 400 + 600 + 1000}{5} \\ &= \frac{2500}{5}\end{aligned}$$

$$= 500$$

Standard deviation

x	$x - \overset{(\mu)}{\mu}$ $x - 500$	$\overset{(\mu)}{(x - 500)}^2$
200	-300	90,000
300	-200	40,000
400	-100	10,000
600	100	10,000
1000	500	250,000

$$\sum (x - \mu)^2 = 400000$$

$$= \frac{400000}{5}$$

$$= 80000$$

$$\sigma = \sqrt{80000}$$

$$\sigma \approx 282.64$$

Z-Score;

$$Z = \frac{x - \mu}{\sigma} = \frac{300 - 500}{282.64}$$

$$Z = \frac{200 - 500}{282.64}$$

$$= -1.06$$

$$= -0.71$$

original value	Z-Score
200	-1.06
300	-0.71
400	-0.35
600	0.35
1000	1.77

5. Suppose a group of 12 sales price record has been sorted as follow; 5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215. partition them into three bins by each of the following methods. (a) equal-frequency partitioning (b) equal-width partitioning (c) clustering

$$\text{no of record} = 12$$

$$\text{no of bin} = 3$$

(a) equal-frequency partition

$$\frac{12}{3} = 4 \text{ record per bin}$$

Bin	values
B ₁	5, 10, 11, 13
B ₂	15, 35, 50, 55
B ₃	72, 92, 204, 215

(b) Equal-width partition

• Min = 5

• Max = 215

Range = $215 - 5 = 210$

Bin width = $\frac{210}{3} = 70$

Form the Bins

• Bin 1 : 5-74

• Bin 2 : 75-144

• Bin 3 : 145-215

Bin	values
B ₁ (5-74)	5, 10, 11, 13, 15, 35, 50, 55, 72
B ₂ (75-144)	92
B ₃ (145-215)	204, 215

(d) Cluster

cluster	values
C ₁	5, 10, 11, 13, 15 ₉₂
C ₂	35, 50, 55, 72
C ₃	204, 215

These group are formed because ;

- C₁ : small value
- C₂ : Medium value
- C₃ : large value

6. Suppose a hospital tested the age & body fat data for 18 randomly selected adult with the following result.

a. calculate the mean, median & standard deviation of age & fat.

b. draw the boxplot for age & fat

c. draw a scatter plot & a q-q plot based on these two variable.

d. find the correlation. Are these two

(c) Cluster

cluster	values
C ₁	5, 10, 11, 13, 15 ₉₂
C ₂	35, 50, 55, 72
C ₃	204, 215

These group are formed because ;

- C₁ : small value
- C₂ : Medium value
- C₃ : large value

6. Suppose a hospital tested the age & body fat data for 18 randomly selected adult with the following result.

a, calculate the mean, median & standard deviation of age & fat.

b, draw the boxplot for age & fat

c, draw a scatter plot & a q-q plot based on these two variable.

d, calculate the correlation. so these two

(d) Cluster

cluster	values
C ₁	5, 10, 11, 13, 15
C ₂	35, 50, 55, 72
C ₃	204, 215

These group are formed because ;

- C₁ : small value
- C₂ : Medium value
- C₃ : large value

6. Suppose a hospital tested the age & body fat data for 18 randomly selected adult with the following result.

a, calculate the mean, median & standard deviation of age & fat.

b, draw the boxplot for age & fat

c, draw a scatter plot & a q-q plot based on these two variable.

d, calculate the correlation. so these two

variables positively & negatively correlated:

Age	Fat
23	9.5
23	26.5
27	7.8
27	17.0
39	31.4
41	25.9
47	27.4
49	27.2
50	31.2
52	34.6
54	42.5
56	28.8
57	33.4
58	30.2
58	34.1
60	32.9
61	41.2

o, Mean, Median & Standard Deviation

	Mean	Median	Standard deviation
Age	46.44	51.00	13.22
Fat	28.78	30.70	9.25

(b) Boxplot

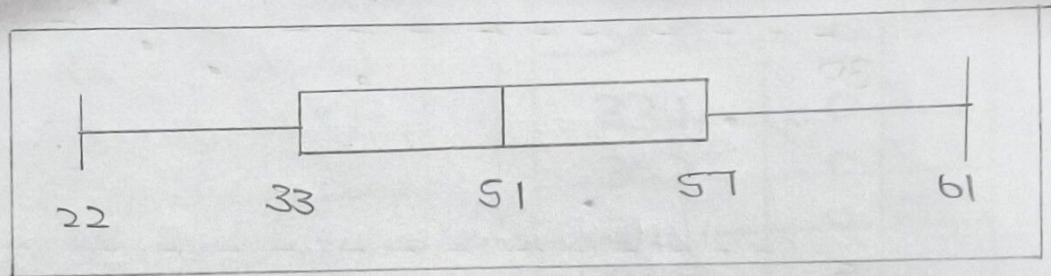
Age;

- Min = 23
- $Q_1 \approx 33$
- Median = 51
- $Q_3 \approx 57$
- Max = 61

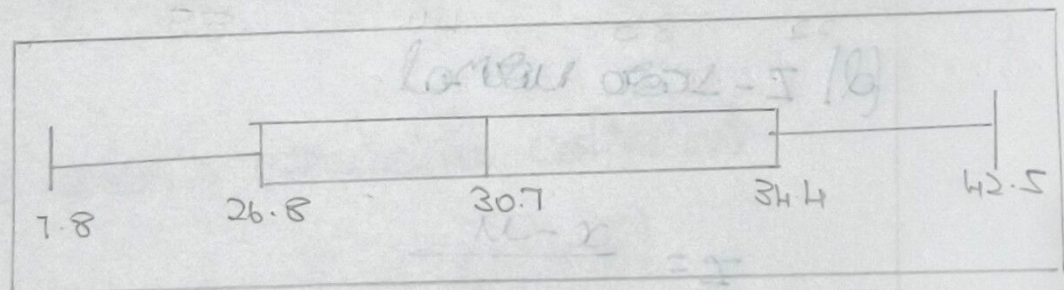
Fat;

- Min = 7.8
- $Q_1 \approx 26.8$
- Median = 30.7
- $Q_3 \approx 34.4$
- Max = 42.5

Age;



Fat;



(c) Scatter plot & Q-Q plot

scatter;

• x-axis - Age

• y-axis - Fat

QQ-plot;

• plot the order

60

45

30

15

0

22

33

44

55

66

(d) z-score normal

$$z = \frac{x - \mu}{\sigma}$$

Age

Age	Z-Score
23	-1.77
23	-1.77
27	-1.47
27	-1.47
39	-0.56
41	-0.41
47	0.04
49	0.19
50	0.27
52	0.42
54	0.57
54	

Foot

Foot	Z-Score
9.5	-2.08
26.5	-0.38
7.8	-3.37
17.8	-1.14
31.4	0.38
25.9	-0.81
27.4	-0.41
27.2	-0.41
31.2	0.38
34.6	